

Bazı istatistiklerin örneklem Dağılımı

örnekleme yer ala
r.d.lerinin bir fonksiyonu. $\rightarrow$ parametre içermez.

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Kitle $N(\mu, \sigma^2)$ dağılımına sahip olsun.

$x_1, x_2, \dots, x_n$ r. d. olsun.

$$\bar{x} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

$$\frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

Kitle dağılımı bilinmiyor.

$$X_1, X_2, \dots, X_n$$

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \quad \text{ANCAK} \quad \begin{matrix} n \geq 30 \\ n \rightarrow \infty \end{matrix} \quad \text{MLT}$$

Teorem

$X_1, X_2, \dots, X_n \sim N(\mu, \sigma^2)$ dağılımından n brlik r. ö. olsun.

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \text{ olur.}$$

ispat

$$M_{X_1}(t) = M_{X_2}(t) = \dots = M_{X_n}(t)$$

$$M_{X_i}(t) = e^{\mu t + \frac{1}{2} t^2 \sigma^2}$$

$$M_{\bar{x}}(t) = \prod_{i=1}^n M_{x_i}(t/n)$$

$$= \underbrace{e^{\mu \frac{t}{n} + \frac{1}{2} \frac{t^2}{n^2} \sigma^2} \dots e^{\mu \frac{t}{n} + \frac{1}{2} \frac{t^2}{n^2} \sigma^2}}_{n \text{ tane}}$$

$$M_{\bar{x}}(t) = e^{\mu t + \frac{1}{2} t^2 \frac{\sigma^2}{n}} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

Ki-kare Dağılımı

Gamma Dağılımından türetilen bir dağılımdır.

$$\text{Gamma: } \frac{1}{\lambda^n \Gamma(n)} x^{n-1} \cdot e^{-x/\lambda}, \quad x > 0$$

$$\text{Gamma fonu: } \int_0^{\infty} x^{n-1} e^{-x} dx = \Gamma(n)$$

n tam sayı $\Gamma(n) = (n-1)!$

$$M_x(t) = (1 - \lambda t)^{-n}$$

$\lambda = 2$, $n = \nu/2$ olarak alınsın.

$$f(x) = \frac{1}{2^{\nu/2} \Gamma(\nu/2)} x^{\nu/2} e^{-x/2}, \quad x > 0 \quad \sim \chi^2_{\nu}$$

$$M_x(t) = (1 - 2t)^{-\nu/2}$$

$$E(X) = \nu, \quad \nu(x) = 2\nu$$

$$\left. \frac{dM_x(t)}{dt} \right|_{t=0} \Rightarrow \left. -\frac{\nu}{2} (1-2t)^{-\frac{\nu}{2}-1} \cdot (-2) \right|_{t=0} = \nu$$

$$\left. \frac{d^2 m_X(t)}{dt^2} \right|_{t=0} \Rightarrow 2v + v^2$$

$$\begin{aligned} V(X) &= E(X^2) - [E(X)]^2 \\ &= 2v + v^2 - v^2 = 2v \end{aligned}$$



Theorem 1: $X \sim N(0,1)$ ise $X^2 \sim \chi^2_1$ olur.

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

std.
sapma

$$\left[\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \right)^2 \right] \sim \chi^2_1$$

$$\left. \begin{array}{l} X \sim P(\lambda) \\ \left[\left(\frac{X - \lambda}{\sqrt{\lambda}} \right)^2 \right] \sim \chi^2_1 \end{array} \right\}$$

ispat

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

$$y = x^2 \Rightarrow g(y) = ?$$

$$f^*(x) = \frac{2}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}, \quad x > 0$$

$$g(y) = f^*(x) \left| \frac{dx}{dy} \right|, \quad \begin{matrix} x = h^{-1}(y) \\ x = \sqrt{y} \end{matrix} \quad \text{ve} \quad \frac{dx}{dy} = \frac{1}{2\sqrt{y}}$$

$$g(y) = \frac{2}{\sqrt{2\pi}} e^{-\frac{1}{2}y} \frac{1}{2\sqrt{y}}$$

$$= \frac{1}{2^{1/2} \Gamma(\frac{1}{2})} y^{\frac{1}{2}-1} e^{-y/2}, \quad y > 0$$

MGF $\int du$ ite,

$$m_x(t) = E(e^{x^2 t}) = \int_{-\infty}^{\infty} e^{x^2 t} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} x^2} dx$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} x^2 (1-2t)} dx$$

$$= \int_{-\infty}^{\infty} N(\mu, \sigma^2) dx = 1$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{1}{2\sigma^2} (x-\mu)^2} dx = 1, \quad \sigma^2 = (1-2t)^{-1}, \quad \mu = 0$$

$$\rightarrow (1-2t)^{-1/2} \underbrace{\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi} (1-2t)^{-1/2}} e^{-\frac{1}{2} x^2 \frac{1}{(1-2t)^{-1}}} dx}_{N(0, (1-2t)^{-1}) = 1}$$



Teorem 2 : $X_1, X_2, \dots, X_n \sim N(0, 1)$ dağılımına sahip bağımsız r.d.leri olsun.

$$Y = \sum_{i=1}^n X_i^2 \sim \chi_n^2$$

$$X_1^2 \sim \chi_1^2 \Rightarrow m_{X_1}(t) = (1-2t)^{-1/2}$$

$$X_2^2 \sim \chi_1^2 \Rightarrow m_{X_2}(t) = (1-2t)^{-1/2}$$

$$\vdots$$
$$X_n^2 \sim \chi_1^2 \Rightarrow m_{X_n}(t) = (1-2t)^{-1/2}$$

$$m_Y(t) = \prod_{i=1}^n m_{X_i^2}(t) = (1-2t)^{-n/2} \sim \chi_n^2$$



Teorem 3 :

$$X_1, X_2, \dots, X_n \sim N(\mu, \sigma^2)$$

$$\frac{X_i - \mu}{\sigma} \sim N(0, 1)$$

$$\left[\left(\frac{X_i - \mu}{\sigma} \right)^2 \right] \sim \chi^2_1$$

$$\sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)^2 \sim \chi^2_n$$

$$X_1 \sim \chi^2_{\nu_1}, X_2 \sim \chi^2_{\nu_2}, \dots, X_n \sim \chi^2_{\nu_n}$$

$$Y = \sum_{i=1}^n X_i \sim \chi^2_{\nu}; \quad \nu = \sum_{i=1}^n \nu_i$$

ispat: MGF $\bigcup_{i=1}^n$

$$m_{X_i}(t) = (1-2t)^{-\nu_i/2}, \quad i=1, 2, \dots, n$$

$$m_Y(t) = \prod_{i=1}^n m_{X_i}(t) = (1-2t)^{-\sum \nu_i/2} \sim \chi^2_{\sum_{i=1}^n \nu_i}$$

Teorem 4: $X_1 \sim \chi^2_{\nu_1}$, $X_1 + X_2 \sim \chi^2_{\nu}$; X_1, X_2 bağımsız r.d.leri olsun. Buna göre $X_2 \sim \chi^2_{\nu - \nu_1}$ olur.

ispat: MCF yöntemi

$$Y = X_1 + X_2$$

$$M_Y(t) = M_{X_1}(t) M_{X_2}(t)$$

X_1 ve X_2 bağımsız.

$$(1-2t)^{-\nu/2} = (1-2t)^{-\nu_1/2} M_{X_2}(t)$$

$$\Rightarrow M_{X_2}(t) = (1-2t)^{-(\nu - \nu_1)/2}$$

* **teorem 5:** $N(\mu, \sigma^2)$, σ^2 biliniyor, dağılımından n birimlik rō $x_1, x_2, \dots, x_n$ olsun. Örneklem varyansı s^2 olarak verilsin.

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{n-1} \text{ olur.}$$

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$= \frac{\sum (x_i - \bar{x})^2}{\sigma^2} \sim \chi^2_{n-1}$$

ispat

$n=2$ isin x_1, x_2

$$\bar{x} = \frac{1}{2} (x_1 + x_2)$$

$$s^2 = \frac{1}{2-1} \sum_{i=1}^2 (x_i - \bar{x})^2 = (x_1 - \bar{x})^2 + (x_2 - \bar{x})^2$$

$$= \left[x_1 - \frac{1}{2} (x_1 + x_2) \right]^2 + \left[x_2 - \frac{1}{2} (x_1 + x_2) \right]^2$$

$$= \left[\frac{1}{2} (x_1 - x_2) \right]^2 + \left[\frac{1}{2} (x_2 - x_1) \right]^2$$

$$= 2 \left[\frac{1}{2} (x_1 - x_2) \right]^2 = \frac{(x_1 - x_2)^2}{2}$$

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{n-1}$$

$$n=2 \Rightarrow \frac{s^2}{\sigma^2} \sim \chi^2_1$$

$$(x_1 - x_2) \sim N(0, 2\sigma^2)$$

$$\frac{(x_1 - x_2)}{\sqrt{2} \sigma} \sim N(0, 1)$$

$$\frac{(x_1 - x_2)^2}{2\sigma^2} \sim \chi^2_1$$

$$s^2 = \frac{(x_1 - x_2)^2}{2}$$

$$\frac{s^2}{\sigma^2} \Rightarrow \frac{(x_1 - x_2)^2}{2\sigma^2} \sim \chi^2_1 \quad \checkmark$$

Genel durum

$x_1, x_2, \dots, x_n \sim N(\mu, \sigma^2)$ bağımsız r.d. olsun.

$$\frac{x_1 - \mu}{\sigma} \sim N(0, 1)$$

$$\frac{x_2 - \mu}{\sigma} \sim N(0, 1)$$

$\vdots$

$$\frac{x_n - \mu}{\sigma} \sim N(0, 1)$$

Teorem 2'den $\left\{ \sum_{i=1}^n \left(\frac{x_i - \mu}{\sigma} \right)^2 \sim \chi_n^2 \right.$

$$\underbrace{\frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu)^2}_{\chi_n^2} = \frac{1}{\sigma^2} \sum_{i=1}^n [(x_i - \bar{x}) + (\bar{x} - \mu)]^2$$

$$= \frac{1}{\sigma^2} \left[\sum_{i=1}^n (x_i - \bar{x})^2 + n(\bar{x} - \mu)^2 \right]$$

$$= \frac{1}{\sigma^2} \left[\sum_{i=1}^n (x_i - \bar{x})^2 + n(\bar{x} - \mu)^2 \right]$$

$$\chi_n^2 = \frac{(n-1)s^2}{\sigma^2} + \underbrace{\left[\frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \right]^2}_{\chi_1^2}$$

$\bar{x}$ ve s^2 bağımsız.

Bu durumda teorem 4'den $s^2 \sim \chi_{n-1}^2$ olur.

$$\sum_{i=1}^n 2(x_i - \bar{x})(\bar{x} - \mu)$$

$$2(\bar{x} - \mu) \underbrace{\sum_{i=1}^n (x_i - \bar{x})}_{\sum x_i - n\bar{x}}$$

$$\sum x_i - n\bar{x}$$

$$n\bar{x} - n\bar{x} = 0$$

